

WASP-177b

R-2233

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-177_201108 · created 2026-08-15T01:26:31+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459161.6362 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.049
Transit depth (Rp/Rs)^2	2.88 +/- 1.65 [%]
Orbital Inclination (inc)	86.9 +/- 2.85
Airmass coefficient 1 (a1)	0.211 +/- 0.023
Airmass coefficient 2 (a2)	1.29 +/- 1.47
Scatter in the residuals of the lightcurve fit is	10.34 %
Best Comparison Star	#3 - [547, 134]
Optimal Aperture	4.64
Optimal Annulus	10.12
Transit Duration (day)	0.042 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.049 vs 0.1618 (deviation 0.17σ)
Duration fit vs published	0.042 d vs 0.0721 d (deviation 1.77σ)
Mid-time O–C	15.0 min
Depth SNR	3.5
Residual scatter	10.34 %

Light curve

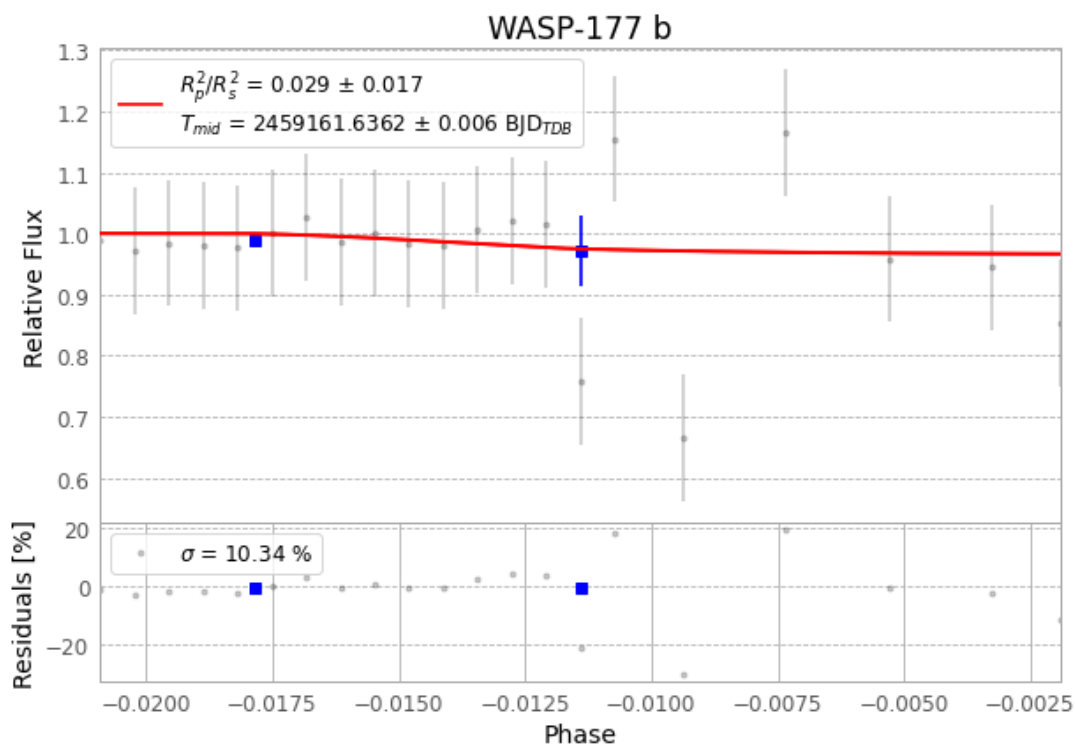


Figure 1 — FinalLightCurve_WASP-177 b_2020-11-07.png (SHA-256 c3fb6152f16491f0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0481d798520c5b0a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

30 FITS frames (20 MB), WASP-177201108013912.FITS → WASP-177201108030613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-201108.log (cc9e6e8183e6...), FinalLightCurve_WASP-177 b_2020-11-07.png (c3fb6152f164...), FinalLightCurve_WASP-177 b_2020-11-07.csv (a4aed7bf42c6...)

Compute resources

Wall time 169.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 01:26Z.